Climate CHANGE per interval − four ponds through the Holocene, survey plots in the Anthropocene

Holocene = UNCORRECTED CHELSA−TraCE21k (the series the analysis uses), 500−yr rolling mean across the 5 moving−window offsets, which are drawn faint behind.
Right of the dotted line: Anthropocene change across survey plots (present − past), one box per site. All paces are PER CENTURY at both ends.

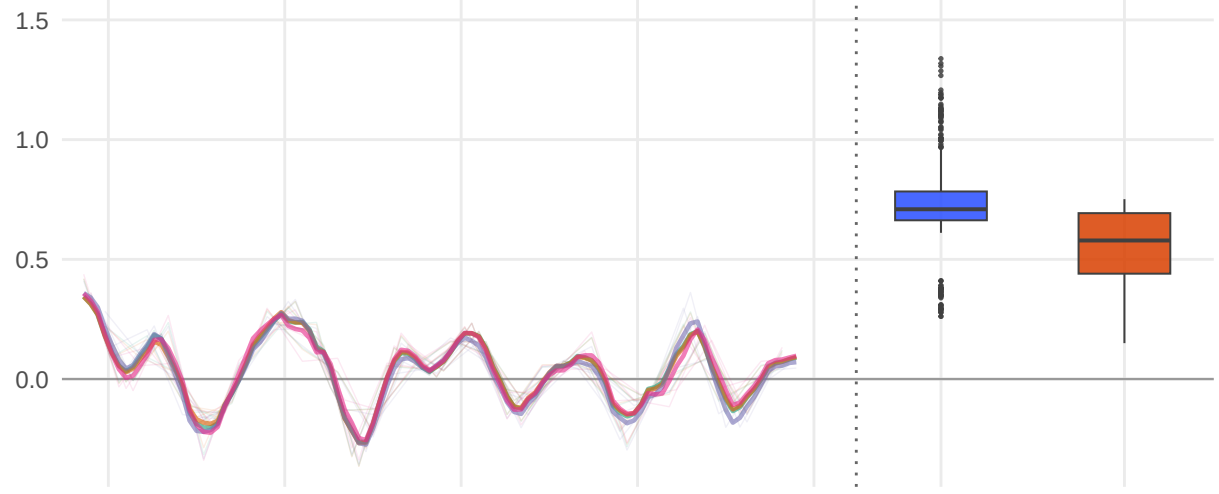
Site (Anthropocene): MORA, NOCA — Pond (Holocene): JayBath, LogWallow, PantherPotholes, ReflectionPond

MAT

degC per century

	min	mean	max
JayBath	−0.072	0.007	0.084
LogWallow	−0.073	0.007	0.084
PantherPotholes	−0.071	0.005	0.082
ReflectionPond	−0.068	0.007	0.087

MORA	0.952	2.474	4.249
NOCA	0.545	1.932	2.636

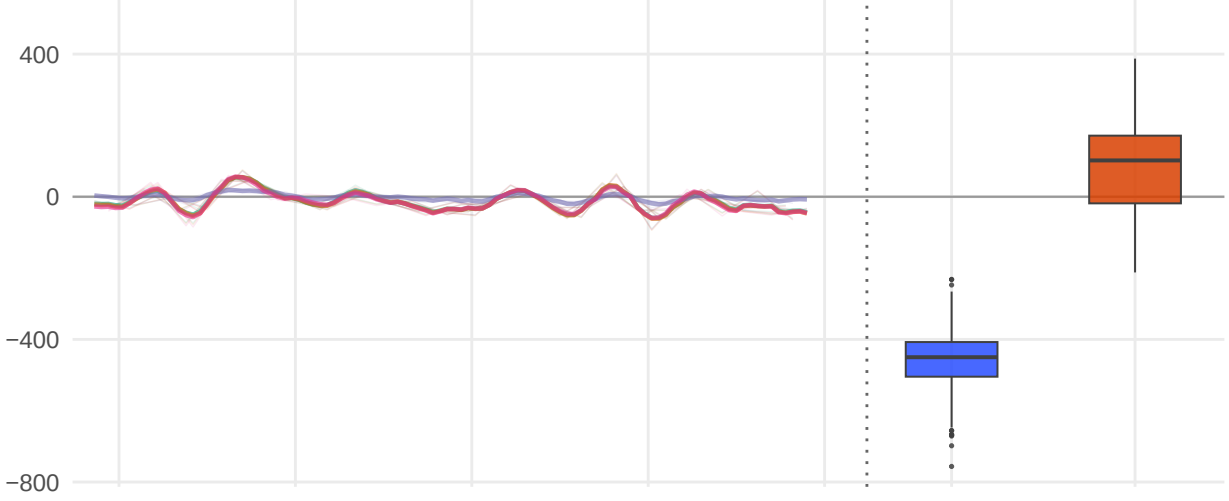


AP

mm per century

	min	mean	max
JayBath	−18.1	−2.528	14.7
LogWallow	−18.9	−2.720	15.1
PantherPotholes	−6.206	−0.404	4.886
ReflectionPond	−18.4	−2.760	14.5

MORA	−2401	−1570	−844
NOCA	−745	321	1359

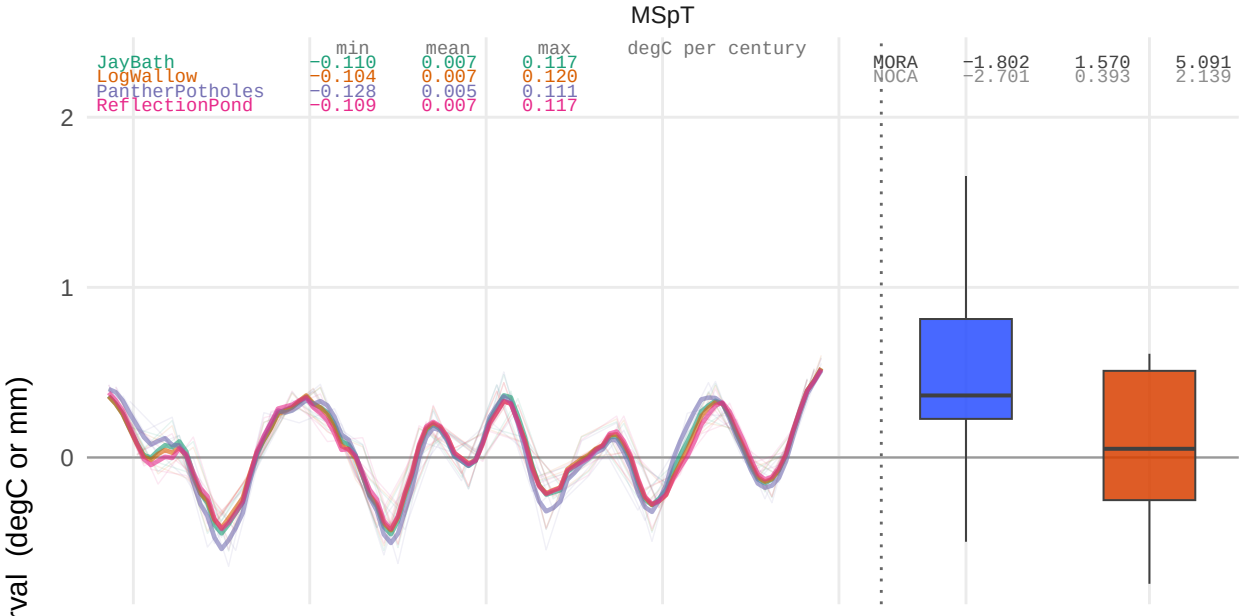


MSpT

degC per century

	min	mean	max
JayBath	−0.110	0.007	0.117
LogWallow	−0.104	0.007	0.120
PantherPotholes	−0.128	0.005	0.111
ReflectionPond	−0.109	0.007	0.117

MORA	−1.802	1.570	5.091
NOCA	−2.701	0.393	2.139

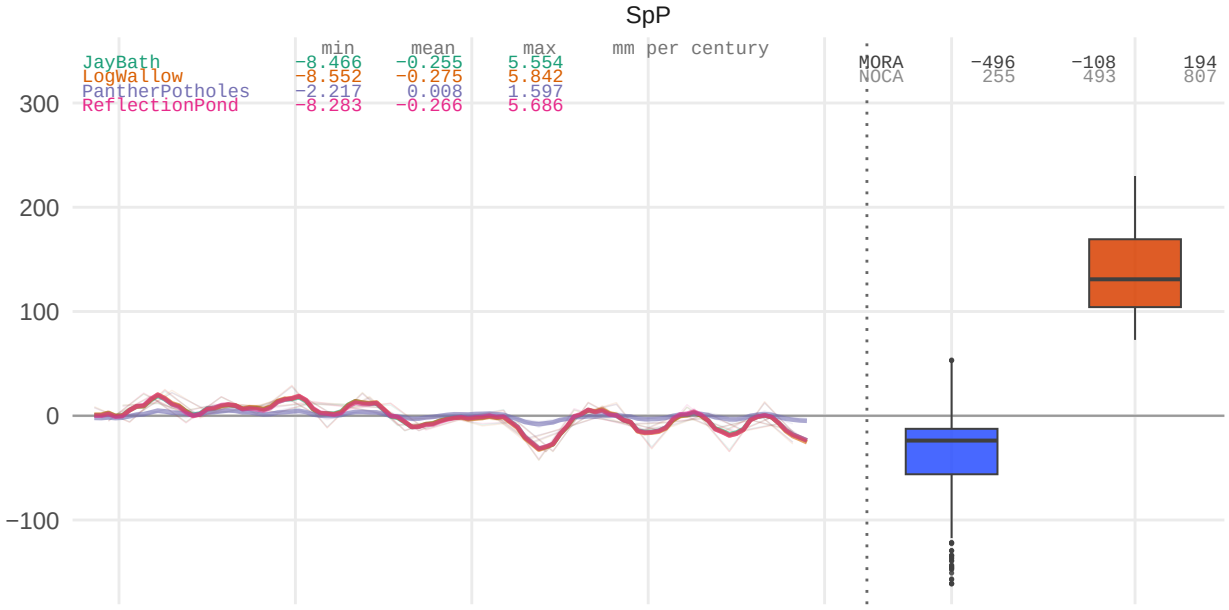


SpP

mm per century

	min	mean	max
JayBath	−8.466	−0.255	5.554
LogWallow	−8.552	−0.275	5.842
PantherPotholes	−2.217	0.008	1.597
ReflectionPond	−8.283	−0.266	5.686

MORA	−496	−108	194
NOCA	255	493	807



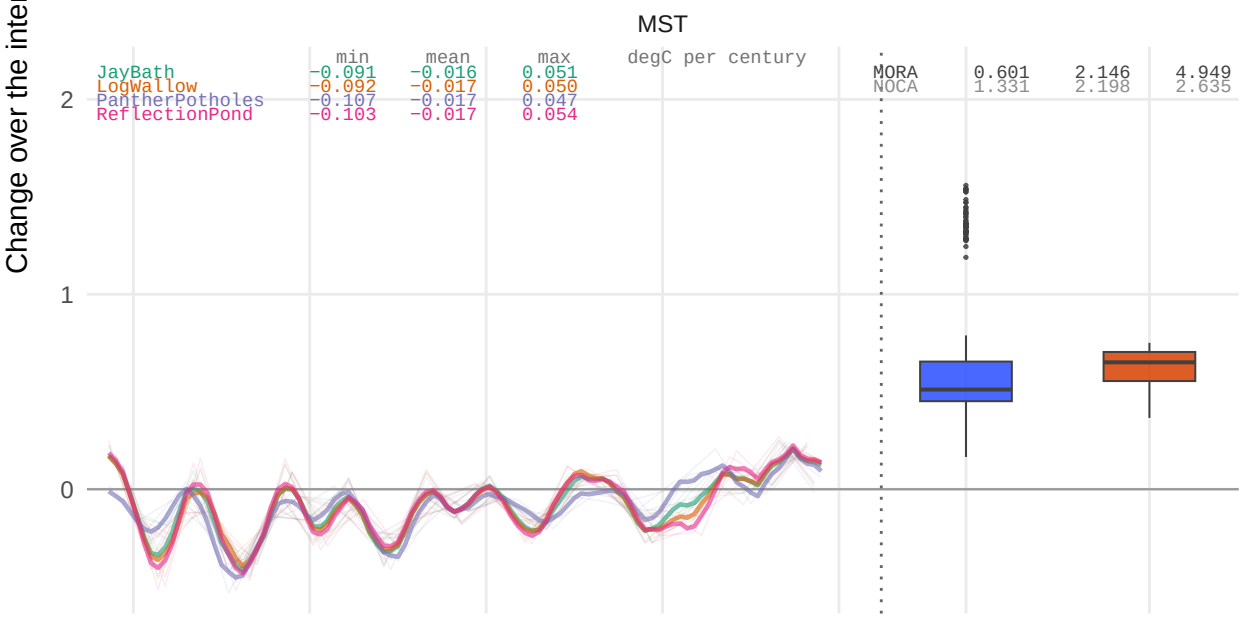
Change over the interval (degC or mm)

MST

degC per century

	min	mean	max
JayBath	−0.091	−0.016	0.051
LogWallow	−0.092	−0.017	0.050
PantherPotholes	−0.107	−0.017	0.047
ReflectionPond	−0.103	−0.017	0.054

MORA	0.601	2.146	4.949
NOCA	1.331	2.198	2.635

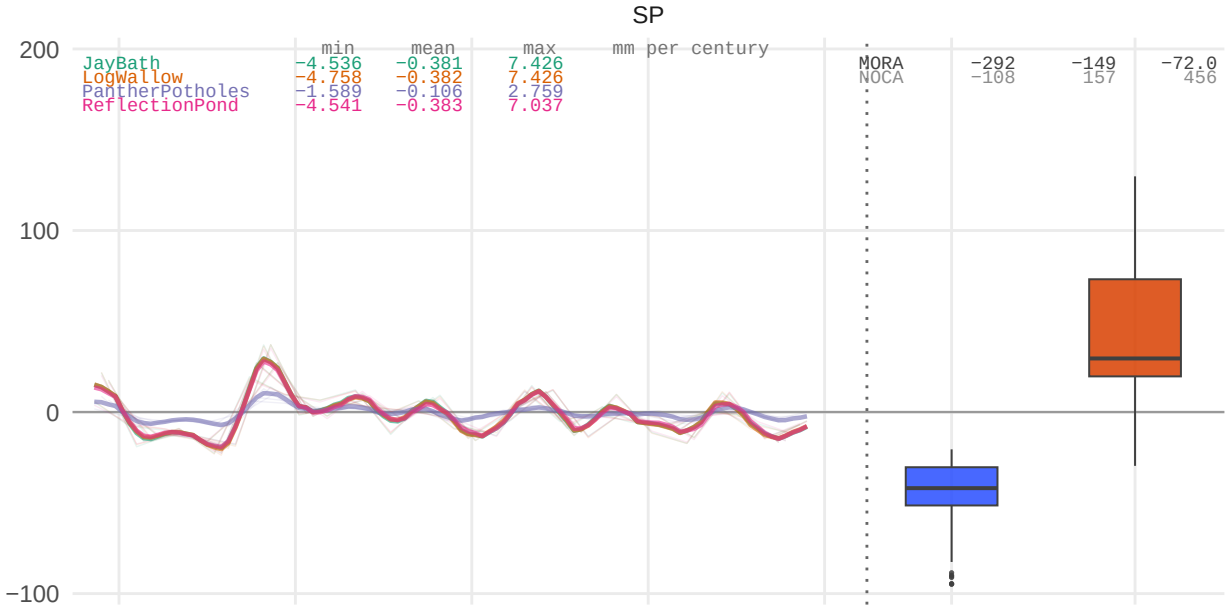


SP

mm per century

	min	mean	max
JayBath	−4.536	−0.381	7.426
LogWallow	−4.758	−0.382	7.426
PantherPotholes	−1.589	−0.106	2.759
ReflectionPond	−4.541	−0.383	7.037

MORA	−292	−149	−72.0
NOCA	−108	157	456

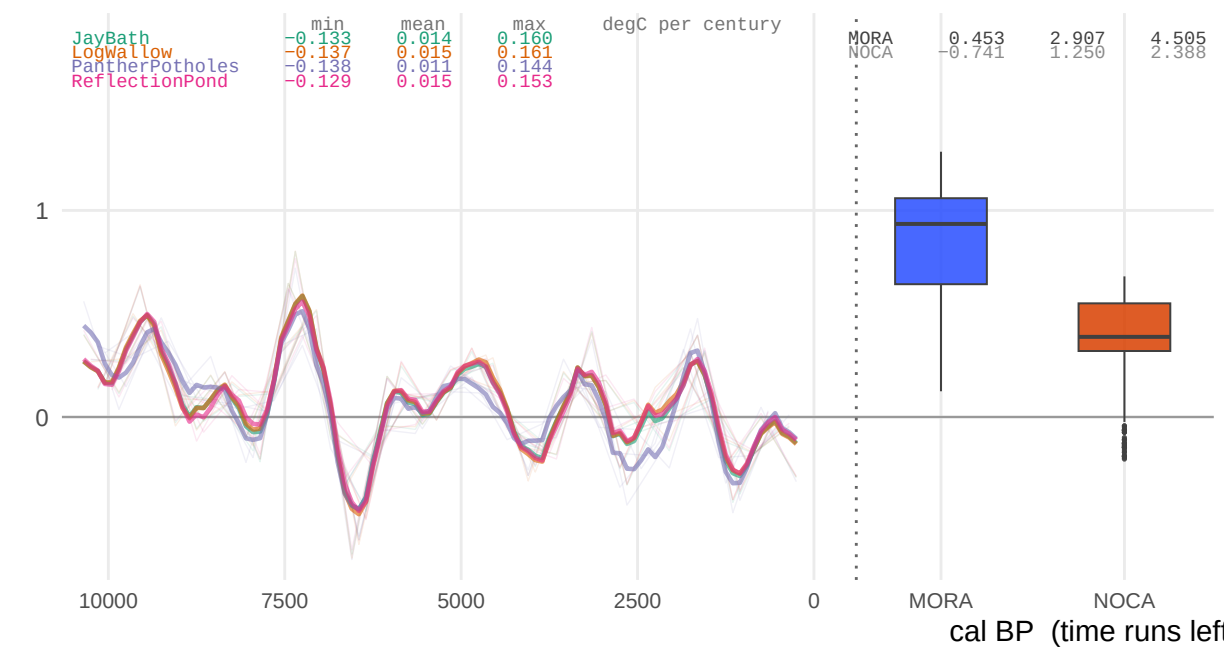


MWT

degC per century

	min	mean	max
JayBath	−0.133	0.014	0.160
LogWallow	−0.137	0.015	0.161
PantherPotholes	−0.138	0.011	0.144
ReflectionPond	−0.129	0.015	0.153

MORA	0.453	2.907	4.505
NOCA	−0.741	1.250	2.388

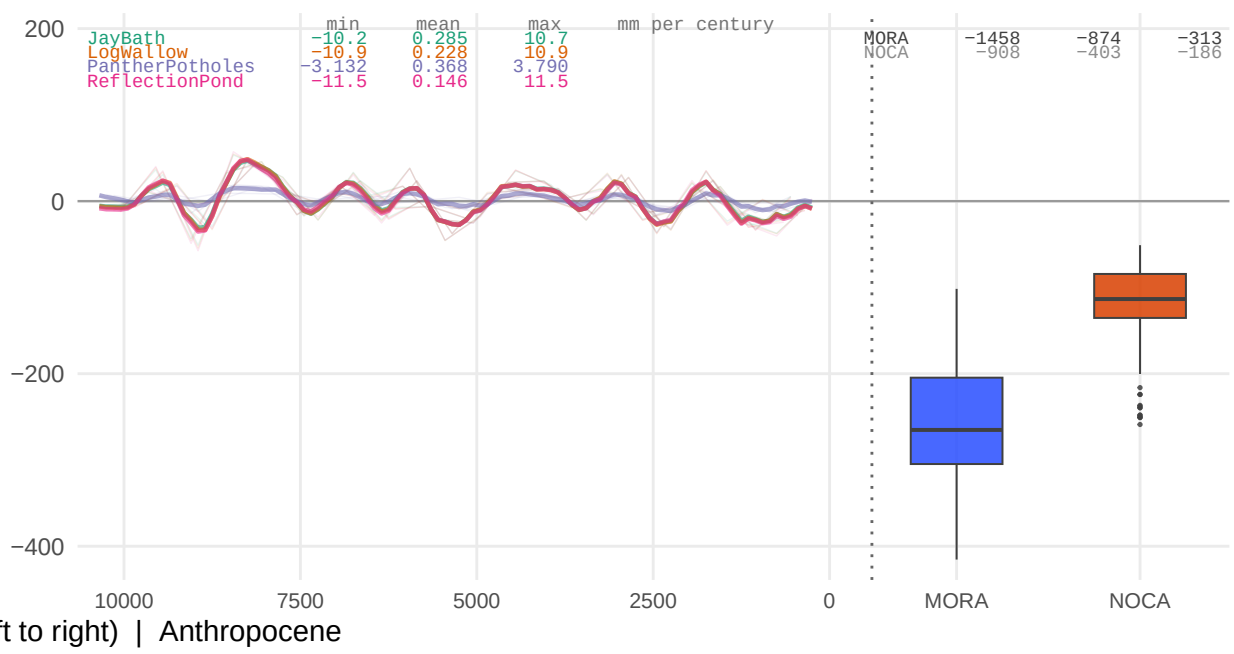


WP

mm per century

	min	mean	max
JayBath	−10.2	0.285	10.7
LogWallow	−10.9	0.228	10.9
PantherPotholes	−3.132	0.368	3.790
ReflectionPond	−11.5	0.146	11.5

MORA	−1458	−874	−313
NOCA	−908	−403	−186

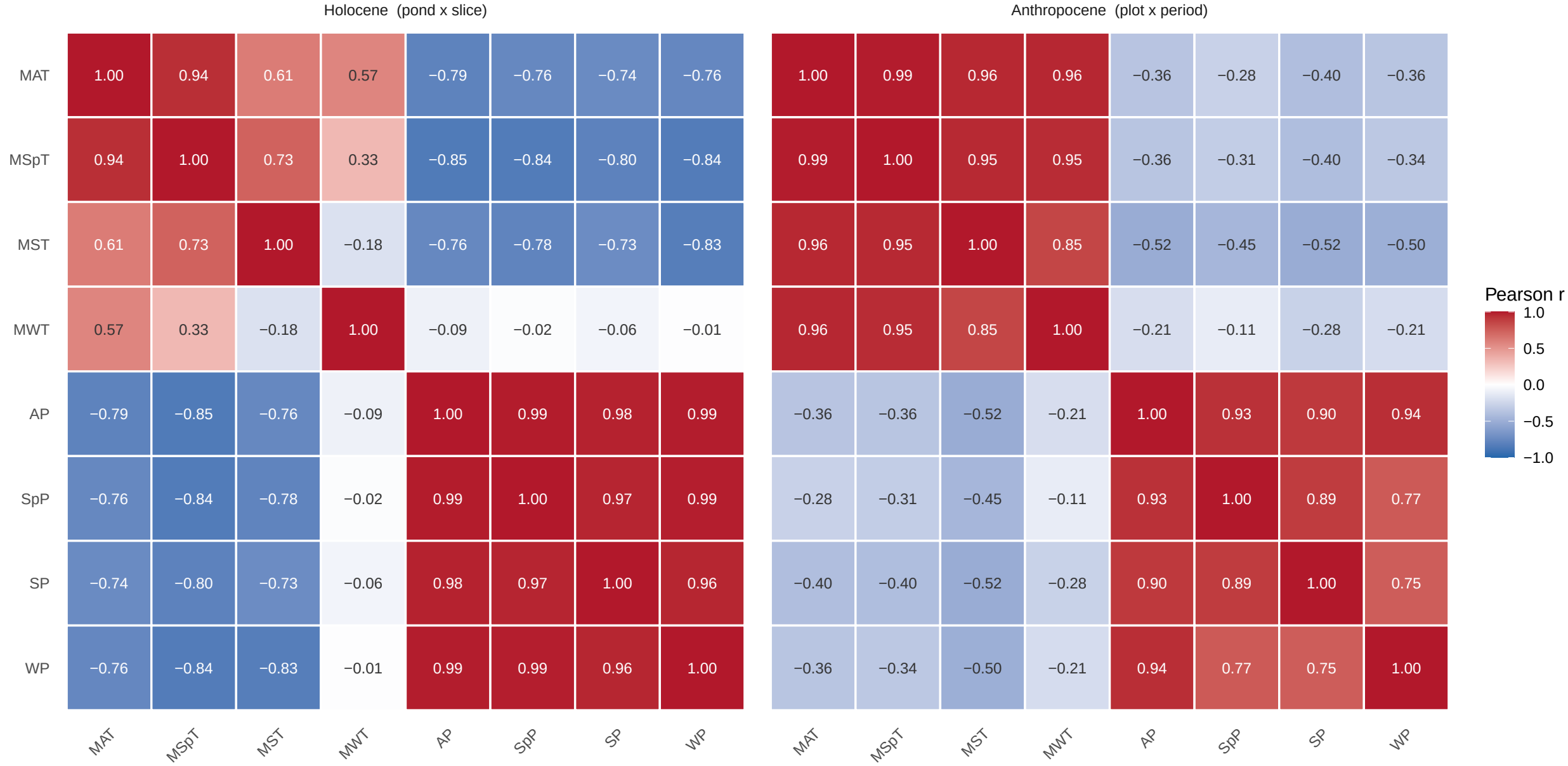


cal BP (time runs left to right) | Anthropocene

Holocene = the FULL TraCE21k record at each pond (all 111 slices, ~100 yr apart), NOT the ages where cores happen to have samples − so the line is the location's climate history, independent of coring. Cores themselves reach JayBath 6841, LogWallow 6808, PantherPotholes 10609, ReflectionPond 6234 cal BP. Intervals differ (500 yr vs ~30 yr): these are CHANGES, not rates − see the

Correlation between climate variables, Holocene vs Anthropocene

The CCA learns the species−climate relationship from the ANTHROPOCENE plots and applies it to HOLOCENE samples.
Contrasting covariance structures can limit transfer; correlations alone do not test predictive validity.



No pair flips sign with |r| > 0.2 in both eras.